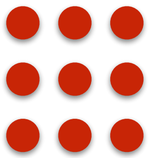
Free Energy Minimization from Nature to LLM

Lei Wang (王磊)
https://wangleiphy.github.io
Institute of Physics, CAS

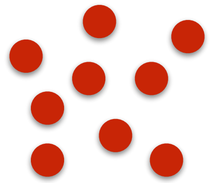
Nature tries to minimize free energy

$$F = E - TS$$

energy



entropy



F is a cost function given by Nature

The SAME cost function for training your favorite LLM

Two phases of LLM training

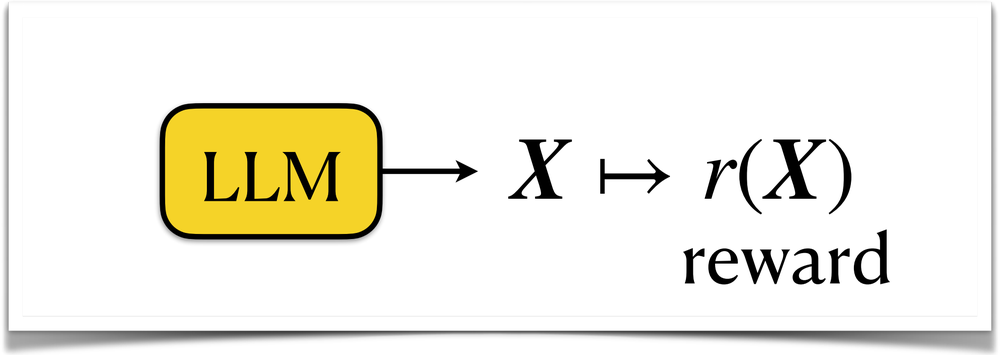
Pre-training



Maximum likelihood estimation on data

$$\mathscr{L} = \mathop{\mathbb{E}}_{X\sim\text{data}} \left[-\ln p(X)\right]$$

Post-training



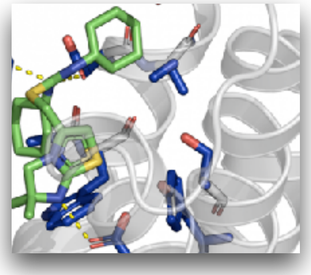
Reinforcement learning from the reward

$$F = \mathop{\mathbb{E}}_{X\sim q(X)} \left[-r(X)\right] + \tau\, \mathbb{KL}\left(q(X)\|p(X)\right)$$

energy entropy

Free energy calculation is useful but difficult

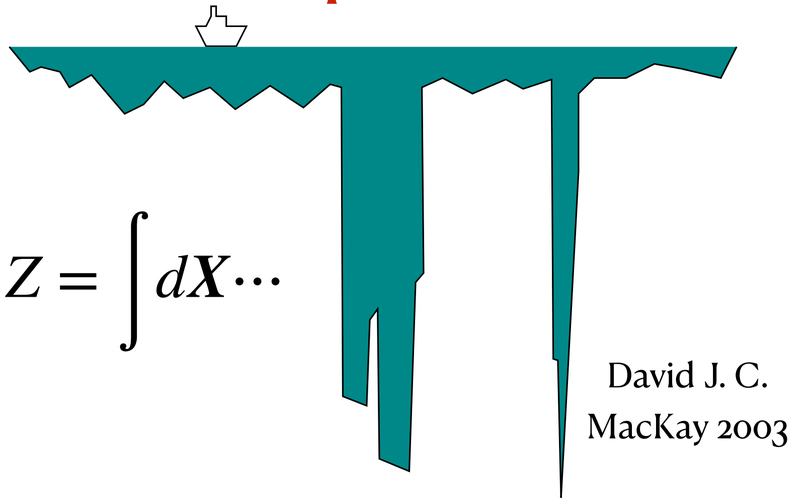
$$F = -k_B T \ln Z$$





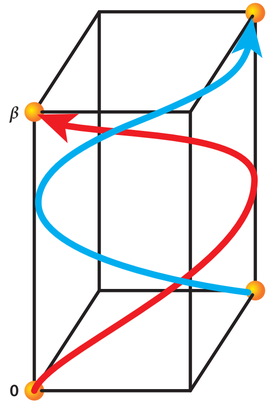

∘ ∘ ∘

"Intractable" partition function



Sign problem in path-integration

$$Z = \int d\tau \, dX \cdots$$

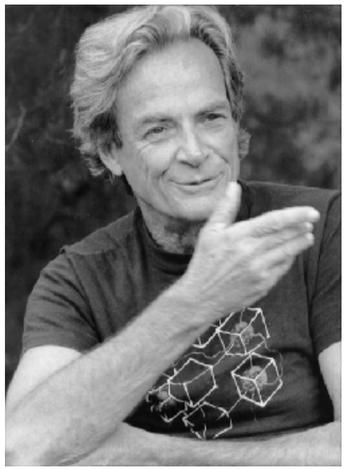


The variational free energy principle

Gibbs–Bogolyubov-Feynman-Delbrück–Molière

$$\min \quad F[\rho] = \mathrm{Tr}(H\rho) + k_B T\, \mathrm{Tr}(\rho \ln \rho) \quad \geq F$$

↓ variational density matrix ↓ energy ↓ entropy



Difficulties in Applying the Variational Principle to Quantum Field Theories[1]

Richard P. Feynman

[1]transcript of Professor Feynman's talk in 1987

1981

15046 citations

Simulating Physics with Computers

Richard P. Feynman

"... it doesn't look so easy."

1987

22 citations

Difficulties in Applying the Variational Principle to Quantum Field Theories[1]

Richard P. Feynman

"...it is no damn good at all!"

Generative AI!

- **Optimization** — **1 Sensitivity to High Frequencies.**
- **Expressibility** — **2 Only Gaussian Trial States**
- **Sampling** — **3 We Still Have To Do a Functional Integral**

Probabilistic modeling with generative AI

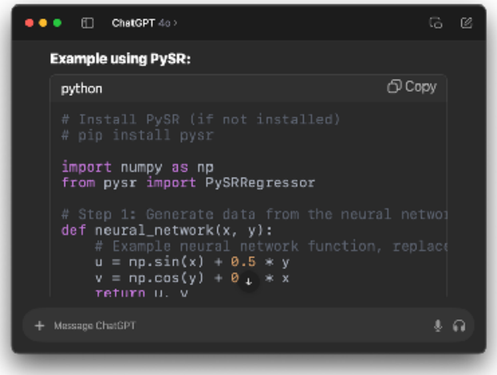
$$p(\boldsymbol{X})$$

pixels, tokens, atoms, ...

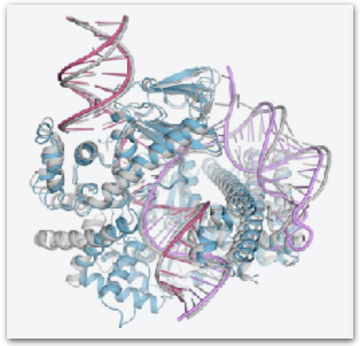
How to express, learn, and sample from a high-dimensional probability distribution ?



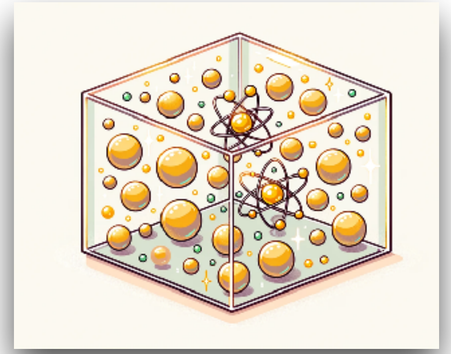
DaLL-E



ChatGPT



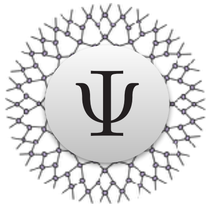
AlphaFold3



this talk

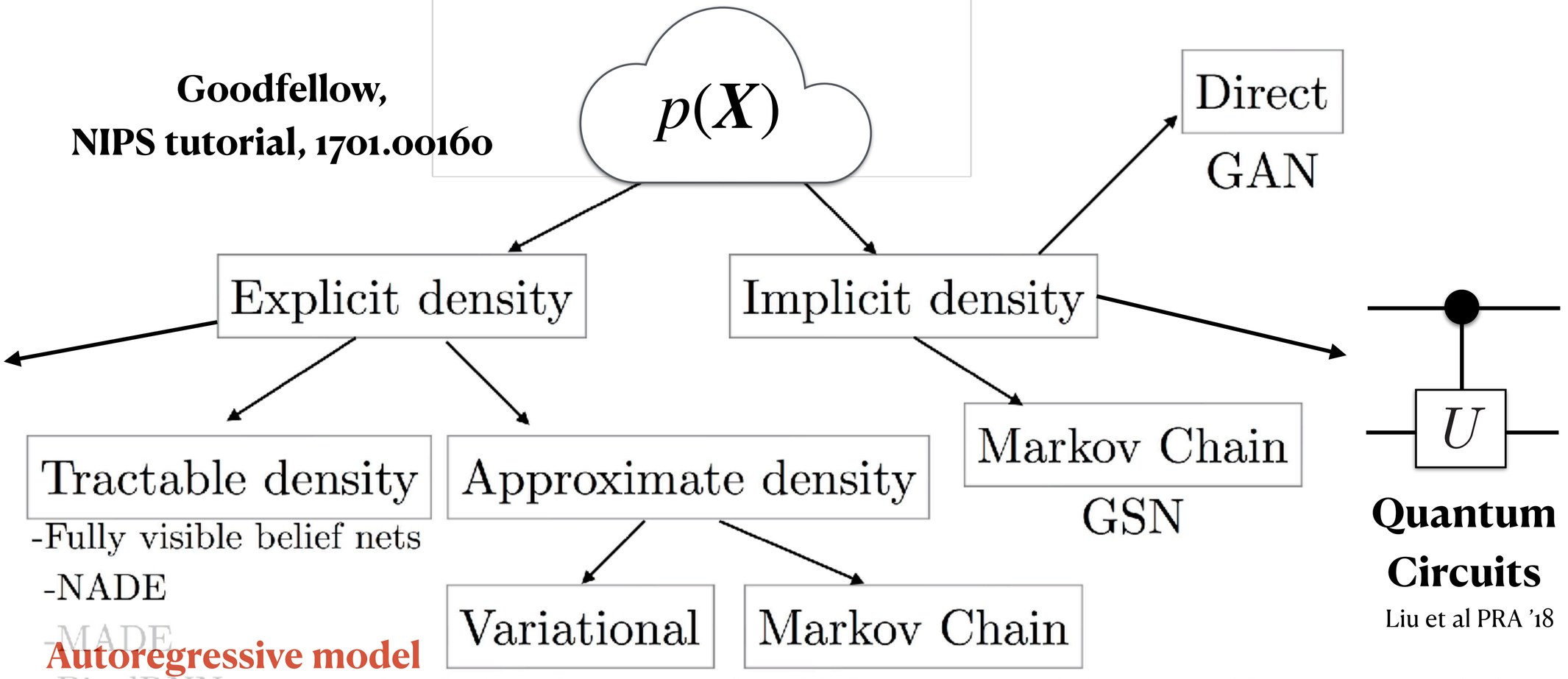
Generative models and their physics genes

Goodfellow,
NIPS tutorial, 1701.00160



Tensor Networks

Han et al, PRX '18

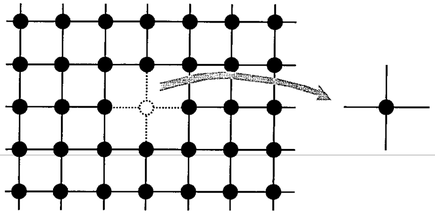


Quantum Circuits

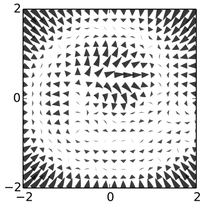
Liu et al PRA '18

+Diffusion models









Tractable generative models

Autoregressive model

$$p(\boldsymbol{X}) = p(x_1)p(x_2 \,|\, x_1)p(x_3 \,|\, x_1, x_2)\cdots$$

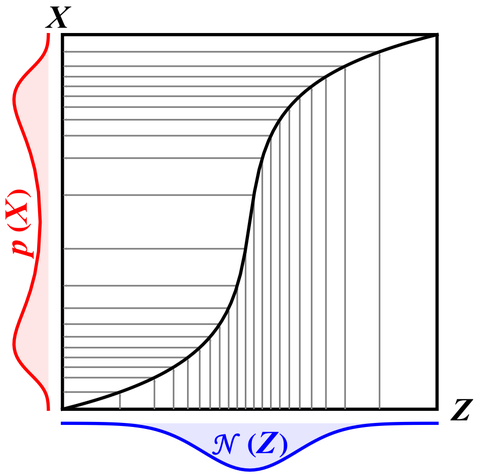


"... the murderer is ___"

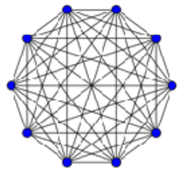
$p(_ \,|\, \ldots)$

Flow model

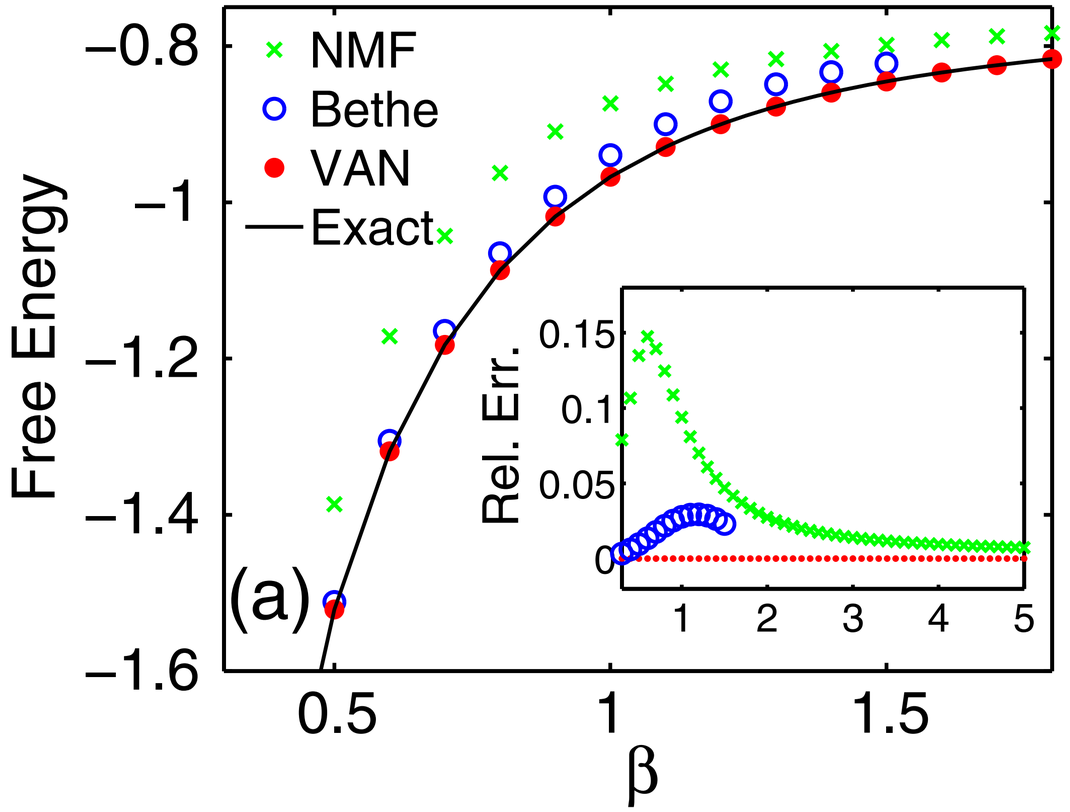
$$p(\boldsymbol{X}) = \mathcal{N}(\boldsymbol{Z}) \left| \det \left(\frac{\partial \boldsymbol{Z}}{\partial \boldsymbol{X}} \right) \right|$$



Variational autoregressive network for statistical mechanics



Sherrington-Kirkpatrick spin glass



Wu, LW, Zhang, PRL '19

Objective function: variational free-energy

$$F = \mathop{\mathbb{E}}_{\boldsymbol{\sigma}\sim p(\boldsymbol{\sigma})} \left[E(\boldsymbol{\sigma}) + k_B T \ln p(\boldsymbol{\sigma}) \right]$$

Variational autoregressive network

$$p(\boldsymbol{\sigma}) = \prod_i p(\sigma_i \,|\, \boldsymbol{\sigma}_{<i})$$

Kikuchi cluster variation method

$$p(\boldsymbol{\sigma}) = \cdots$$

Bethe approximation pairwise interaction

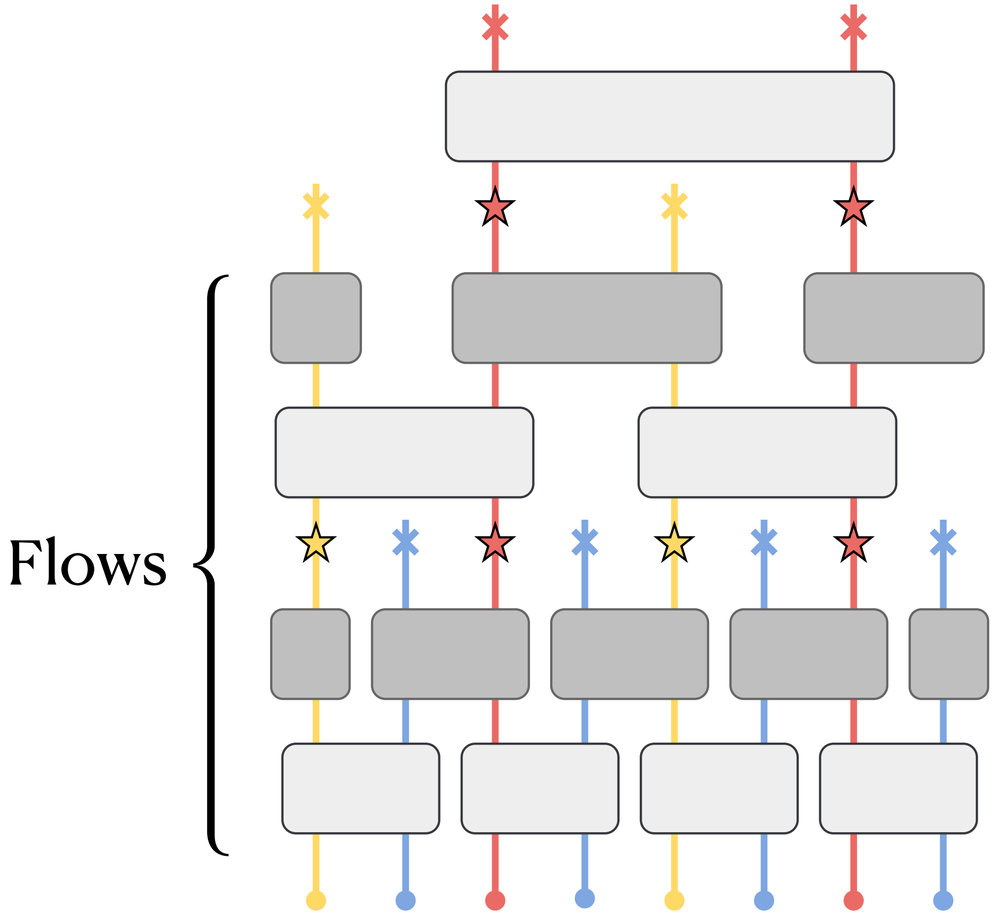
$$p(\boldsymbol{\sigma}) = \prod_i p(\sigma_i) \prod_{(i,j)\in E} \frac{p(\sigma_i, \sigma_j)}{p(\sigma_i) p(\sigma_j)}$$

Naive mean-field factorized probability

$$p(\boldsymbol{\sigma}) = \prod_i p(\sigma_i)$$

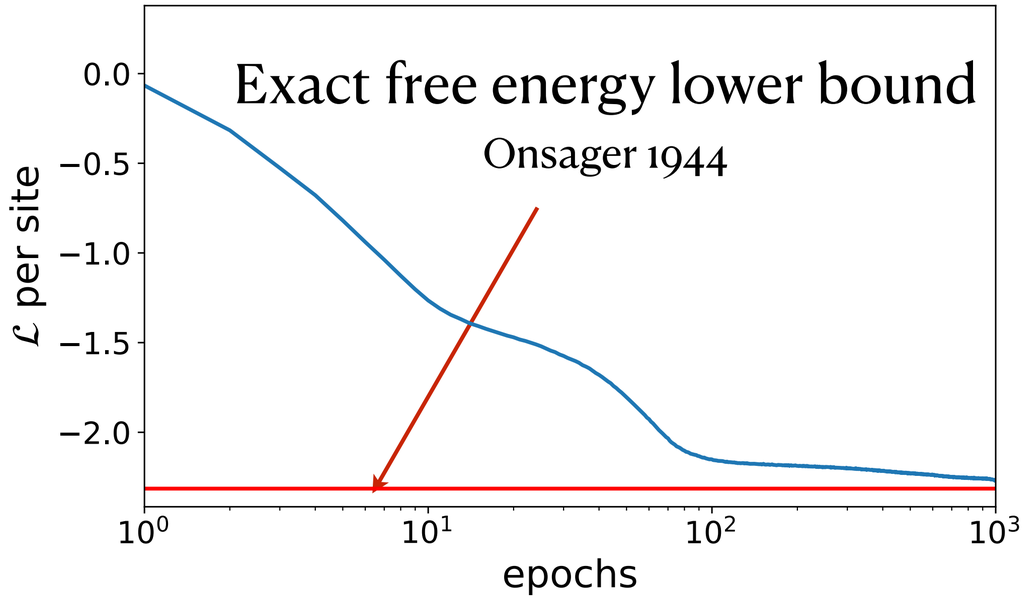
Neural network renormalization group

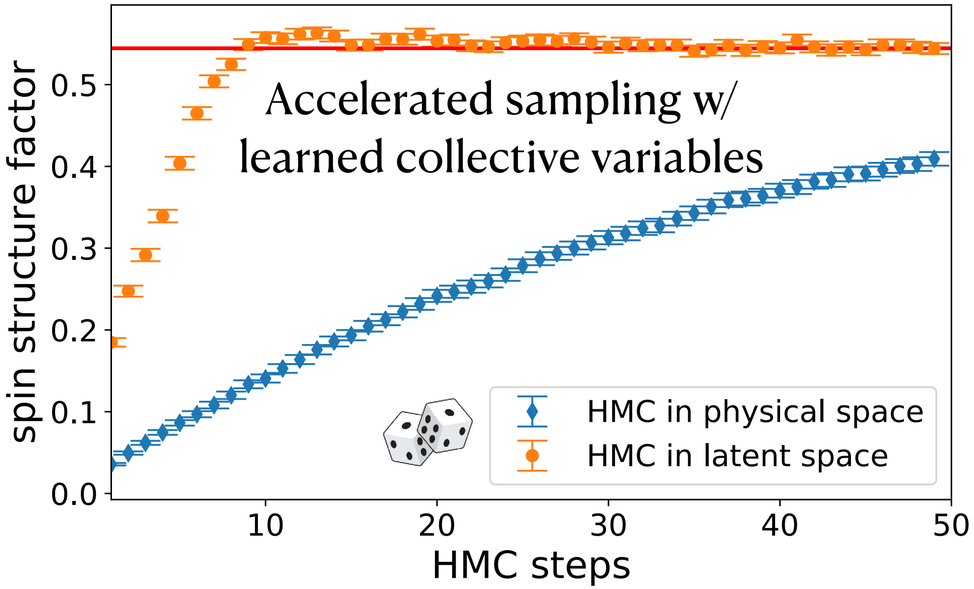
Less correlated collective variables



Correlated physical variables

Li and LW, PRL '18





Neural canonical transformation for variational density matrix

Wu, LW, Zhang, PRL '19

$$\rho = \sum_{\mathbf{\Gamma}} U\,|\mathbf{\Gamma}\rangle\, p_{\mathbf{\Gamma}}\, \langle\mathbf{\Gamma}|\, U^\dagger$$

Cranmer et al, 1904.05903
Xie, Zhang, LW, JML '22

Classical probability $p_{\mathbf{\Gamma}}$

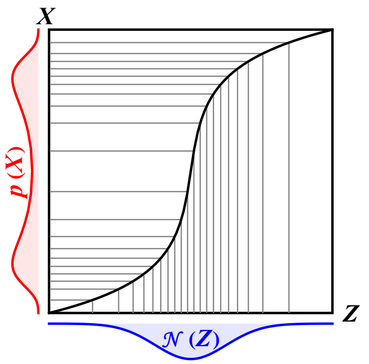
"Base" states
e.g. Plane-wave,
Hartree-Fock,
BCS states

Quantum unitary U



"... the murderer is ___"

$p(_\,|\,\ldots)$



autoregressive model

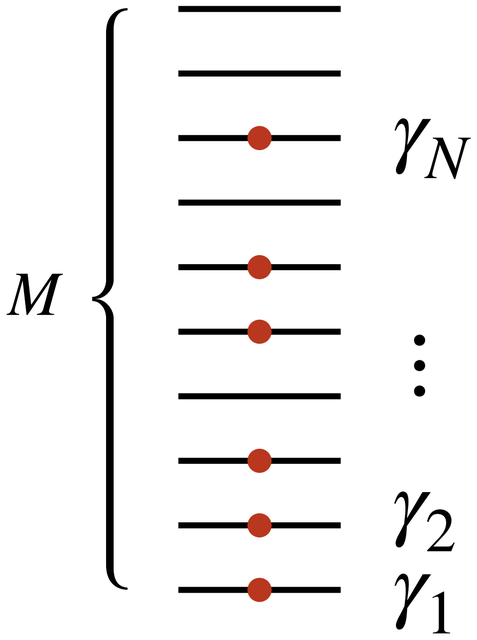
$\sqrt{\text{flow}}$

Autoregressive model for $p_{\mathbf{\Gamma}}$

$$p_{\mathbf{\Gamma}} = p(\gamma_1)p(\gamma_2\,|\,\gamma_1)p(\gamma_3\,|\,\gamma_1,\gamma_2)\cdots$$

Xie, Zhang, LW, SciPost Physics '23

Fermionic distribution



	Physics	**LLM**
N	Number of fermions	Context length
M	Number of orbitals	Vocabulary size
Probability space	$\binom{M}{N}$	M^N
$\ln p_{\mathbf{\Gamma}}$	Landau energy functional	Log-likelihood

"set-of-words"



Pauli exclusion: we are modeling a *set of words* with no repetitions and no order

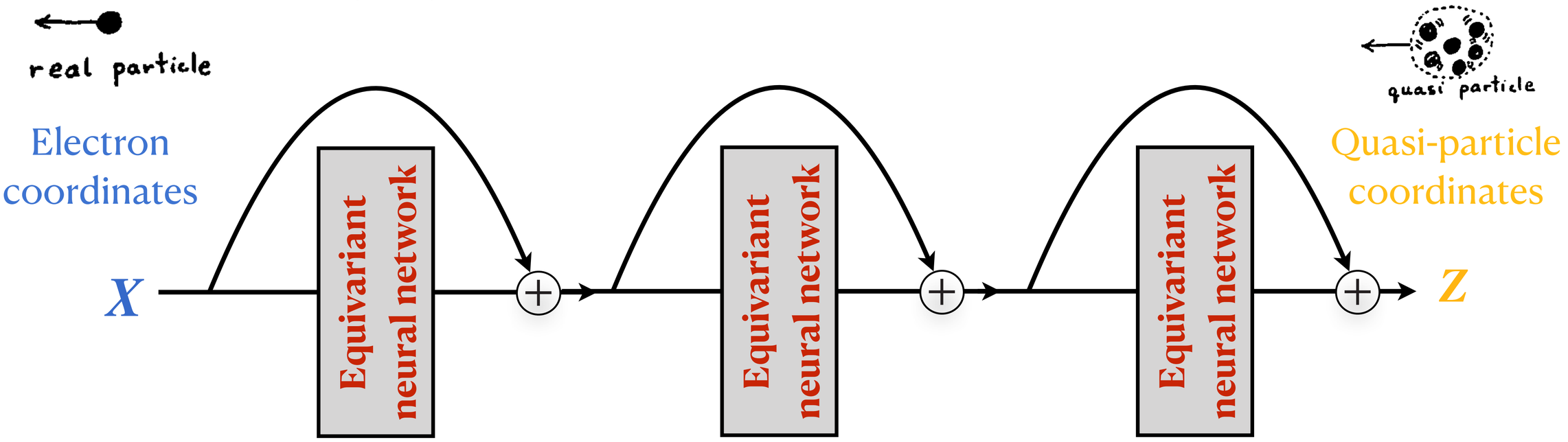
We use masked causal self-attention Vaswani et al 1706.03762; Alternative solution: Hibat-Allah et al, 2002.02973, Barrett et al, 2109.12606

$\sqrt{\text{flow}}$ for unitary

$$\Psi_{\Gamma}(\boldsymbol{X}) = \frac{\det(e^{i\gamma_i \cdot z_j})}{\sqrt{N!}} \cdot \left| \det\left(\frac{\partial \boldsymbol{Z}}{\partial \boldsymbol{X}}\right) \right|^{\frac{1}{2}}$$

Orthonormal many-body states

Xie, Zhang, LW
SciPost Physics '23



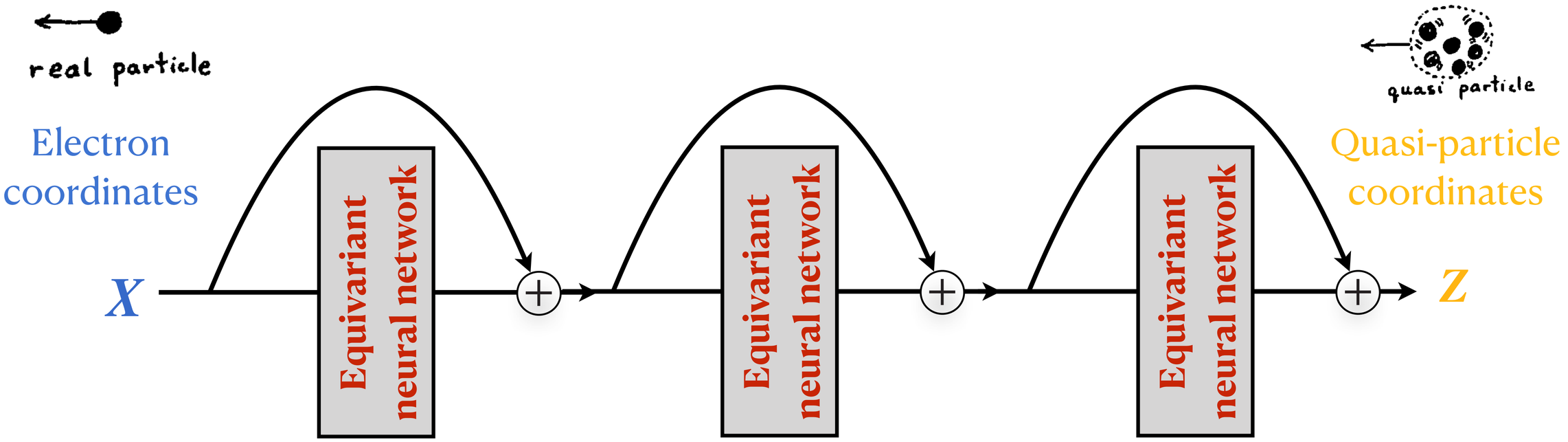
$\boldsymbol{X} \leftrightarrow \boldsymbol{Z}$: Unitary backflow between particle and quasiparticle coordinates

Fermion statistics: permutation equivariant flow we use FermiNet layer Pfau et al, 1909.02487

Feynman's backflow in the deep learning era

$$z_i = x_i + \sum_{j \neq i} \eta(|x_i - x_j|)(x_j - x_i)$$

Feynman & Cohen 1956
wavefunction for liquid Helium



Iterative backflow → Deep residual network → Neural ODE → Continuous normalizing flow→Unitary flow

Taddei et al, PRB '15 E, Commun. Math. Stat. '17, Haber et al 1705.03341, Lu et al 1710.10121, Chen et al, 1806.07366 Xie et al, JML '22

The variational free energy objective

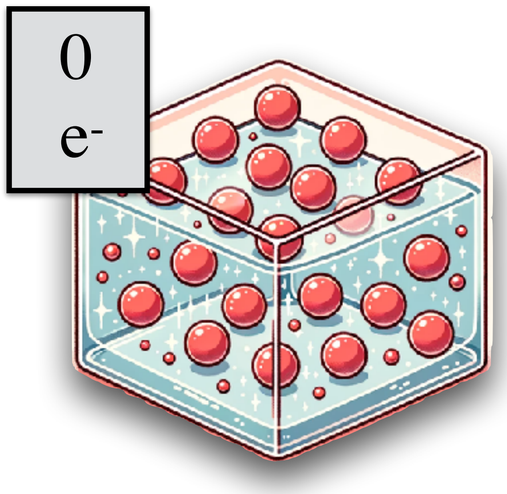
$$F[p_{\mathbf{\Gamma}}, \Psi_{\mathbf{\Gamma}}] = \mathop{\mathbb{E}}_{\mathbf{\Gamma} \sim p_{\mathbf{\Gamma}}} \left[k_B T \ln p_{\mathbf{\Gamma}} + \mathop{\mathbb{E}}_{X \sim \left| \Psi_{\mathbf{\Gamma}}(X) \right|^2} \left[\frac{H \Psi_{\mathbf{\Gamma}}(X)}{\Psi_{\mathbf{\Gamma}}(X)} \right] \right]$$

Boltzmann distribution for energy levels

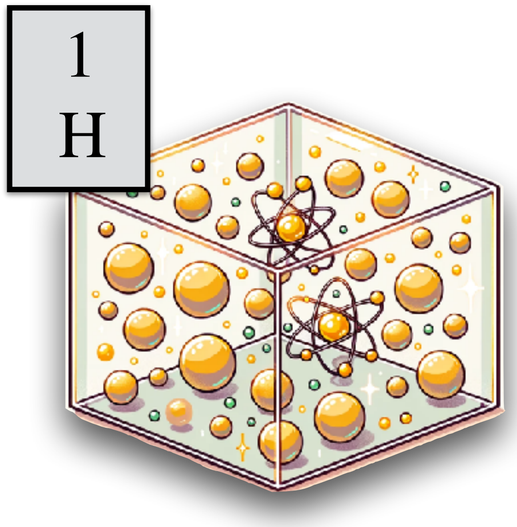
Born probability for coordinates

Jointly optimize exponentially large number of orthonormal many-body states $\mathrm{Tr} \rho \ln \rho = \sum p \ln p$

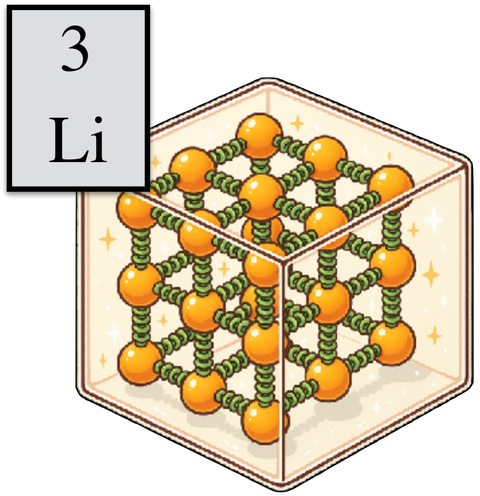
Generative AI in the atomic scale



Electron liquid
phase diagram
JML ’22 and SciPost Physics ’23



Dense hydrogen
equation of state
PRL ’23 and PRL ’26

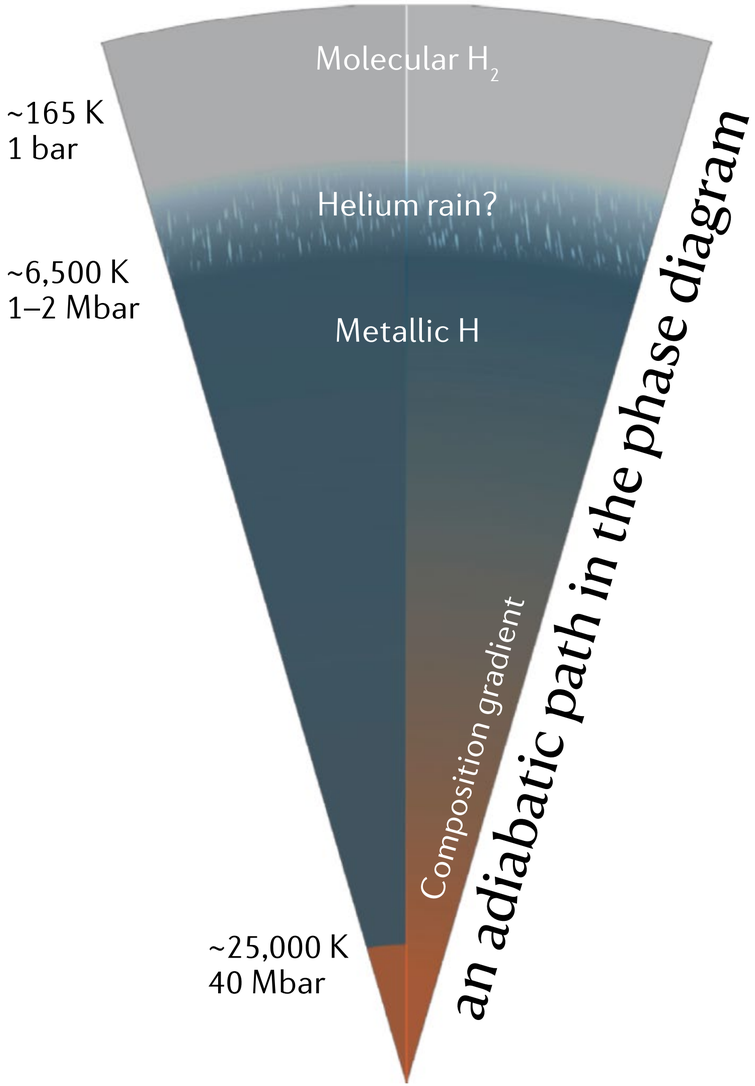


Quantum solids
vibrational structure
JCP ’24 and PRL ’25

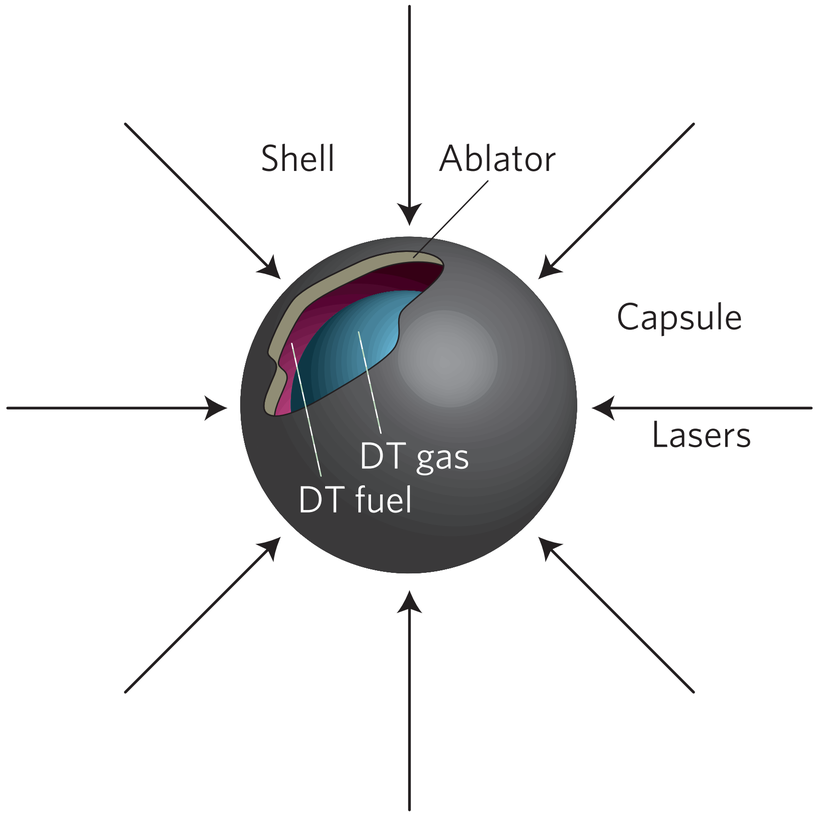
$\min_{\rho} F[\rho]$: minimizing Nature’s cost function with generative AI

Warm dense hydrogen in the sky and in the lab

Jupiter interior



Inertial confinement fusion



Equation-of-state is the input for hydrodynamics simulations

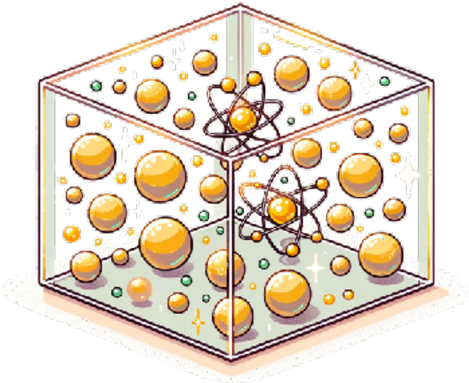
Variational free energy for warm dense hydrogen

Li et al, PRL '26

$$10^{-3} T_F \ll T \lesssim T_F$$

Classical protons

$p(\boldsymbol{S})$



Quantum degenerate electrons

$\Psi_{\boldsymbol{\Gamma}}(\boldsymbol{X}\,|\,\boldsymbol{S})$ and $p_{\boldsymbol{\Gamma}|\boldsymbol{S}}$

proton entropy | electron entropy | internal energy

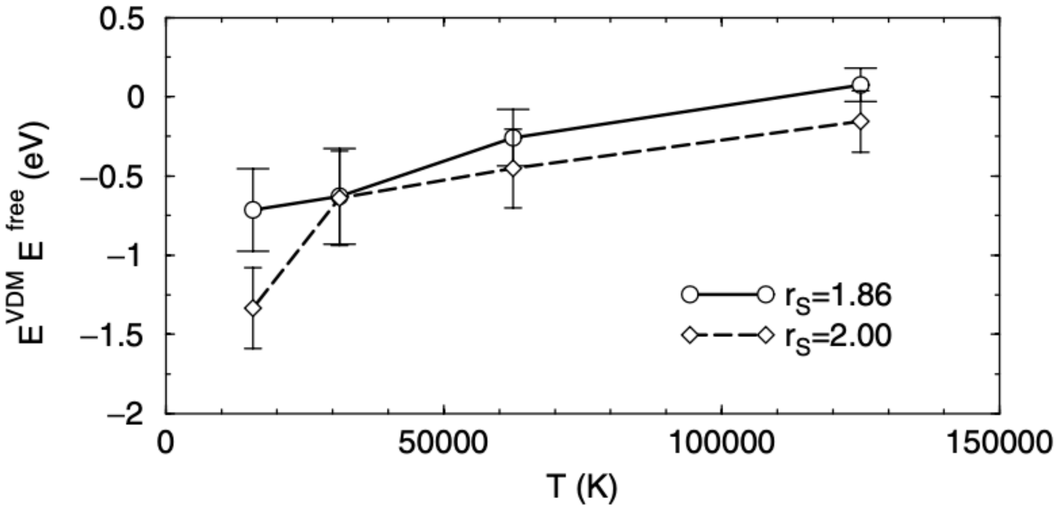
$$F = \mathop{\mathbb{E}}_{\boldsymbol{S}\sim p(\boldsymbol{S})} \left[k_B T \ln p(\boldsymbol{S}) + \mathop{\mathbb{E}}_{\boldsymbol{\Gamma}\sim p_{\boldsymbol{\Gamma}|\boldsymbol{S}}} \left[k_B T \ln p_{\boldsymbol{\Gamma}|\boldsymbol{S}} + \mathop{\mathbb{E}}_{\boldsymbol{X}\sim \left|\Psi_{\boldsymbol{\Gamma}}(\boldsymbol{X}\,|\,\boldsymbol{S})\right|^2} \left[\frac{H\Psi_{\boldsymbol{\Gamma}}(\boldsymbol{X}\,|\,\boldsymbol{S})}{\Psi_{\boldsymbol{\Gamma}}(\boldsymbol{X}\,|\,\boldsymbol{S})} \right] \right] \right]$$



Jointly optimize THREE generative models

A flow for protons, an autoregressive model for energy levels, and a $\sqrt{\text{flow}}$ for electrons

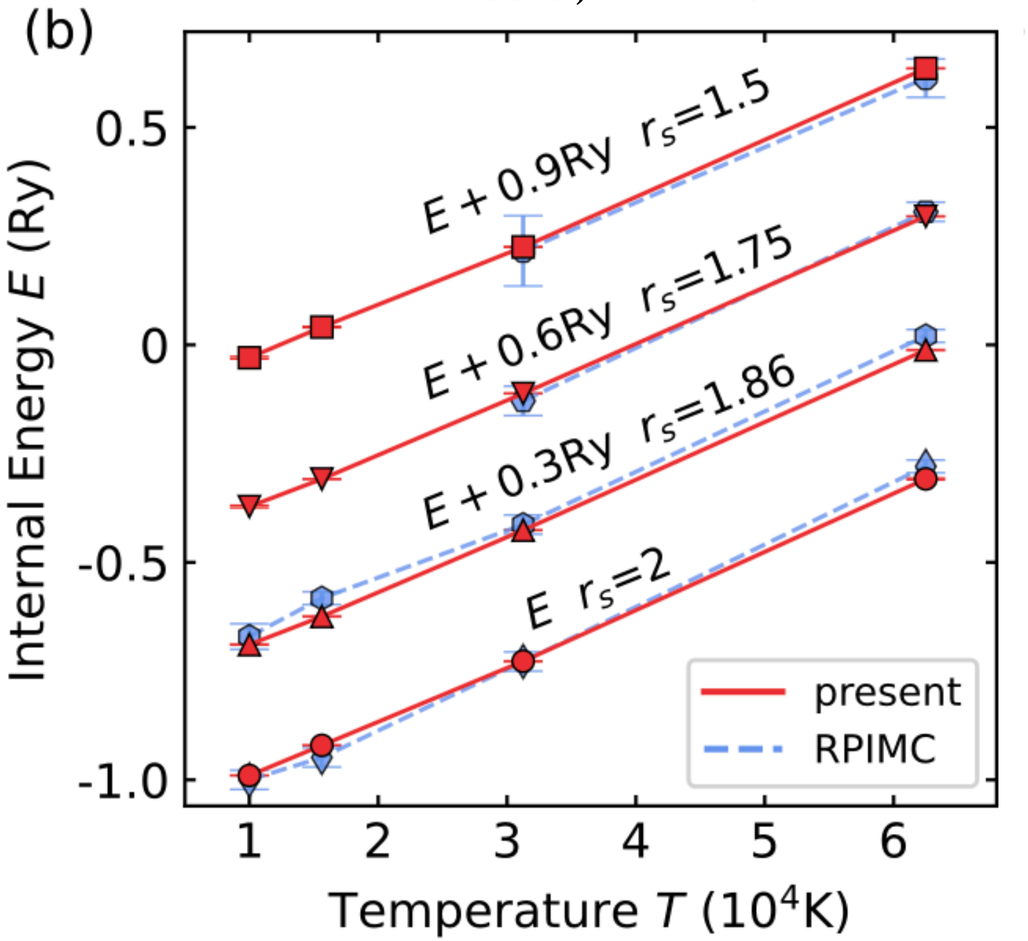
Warm dense hydrogen equation of states



RPIMC results depend on trial density matrices

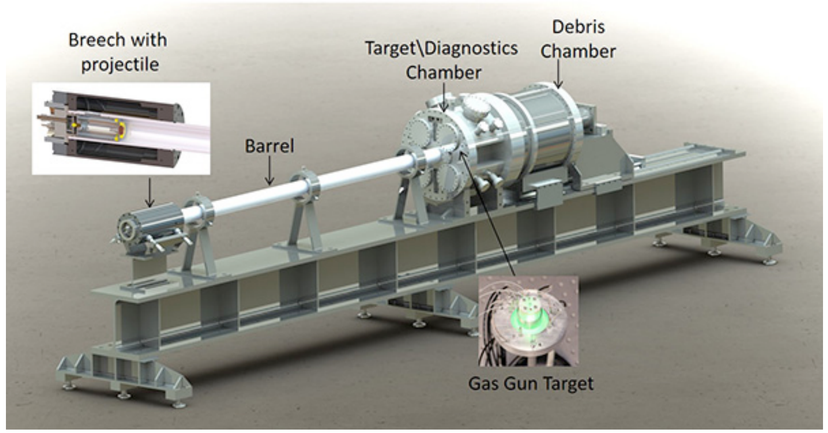
Militzer et al, PRL 2000, PRL 2001

Li et al, PRL '26



Reach lower temperature and higher density compared to RPIMC

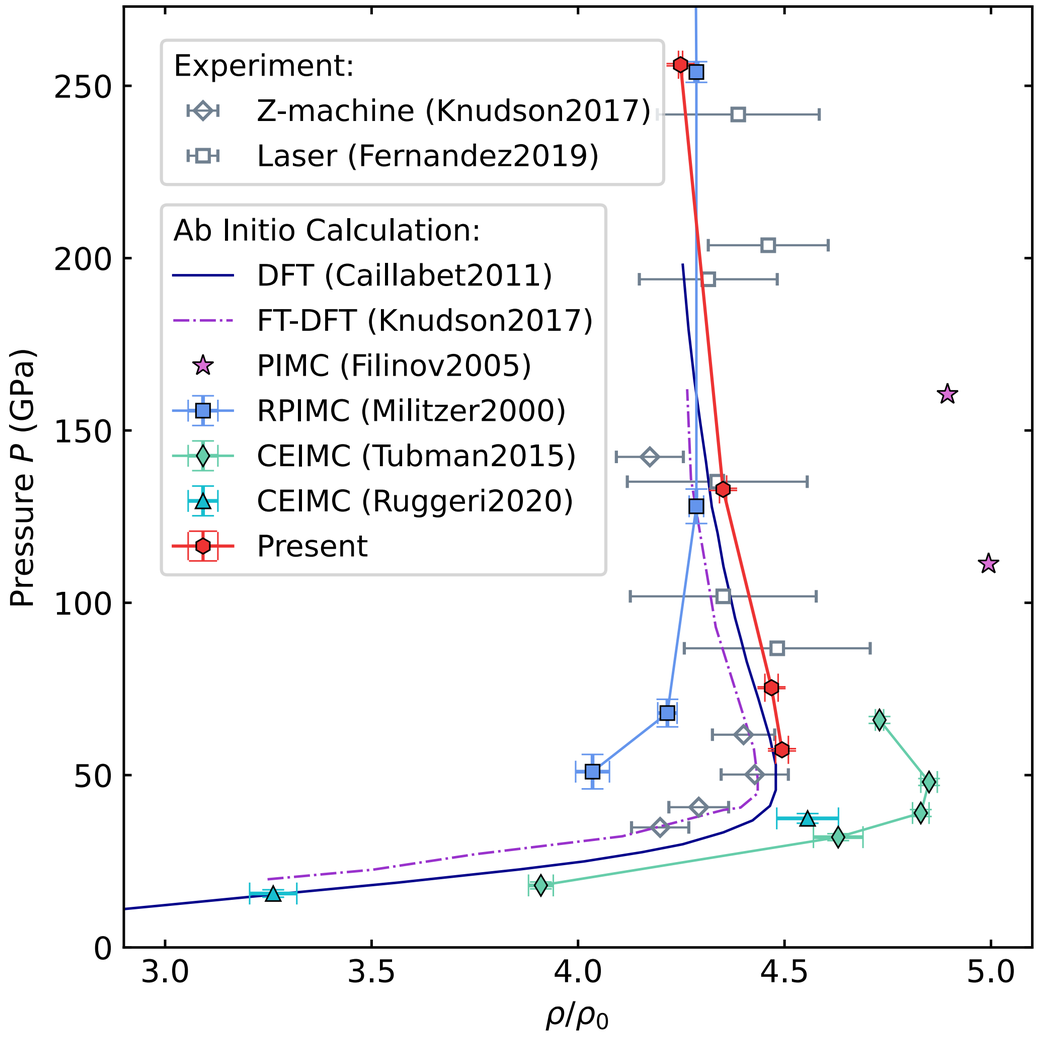
Dense hydrogen Hugoniot curve



Dynamic compression with shock wave

$$E - E_0 + \frac{1}{2}(V - V_0)(p + p_0) = 0$$

Li et al, PRL '26

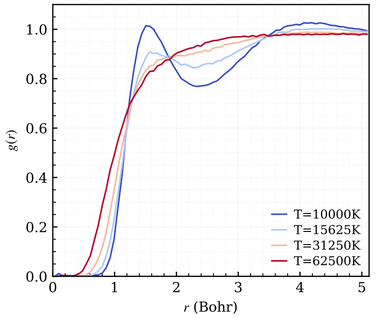


$T = 62500\,\mathrm{K}$

$s = 0.87 k_B$

$T = 10000\,\mathrm{K}$

$s = 0.02 k_B$



The "handshake" 🤝 in dense hydrogen calculation

SAME thing!

1

LLM → $X \mapsto r(X)$

LLM post-training

OpenAI, DeepSeek, ...

2

$$F[\rho] = E - TS$$

Nature's cost function

Xie et al JML '22, SciPost Physics '23...

SAME thing!

1

LLM $\longrightarrow$ $X \mapsto r(X)$

LLM post-training

OpenAI, DeepSeek, ...

2

$$F[\rho] = E - TS$$

Nature's cost function

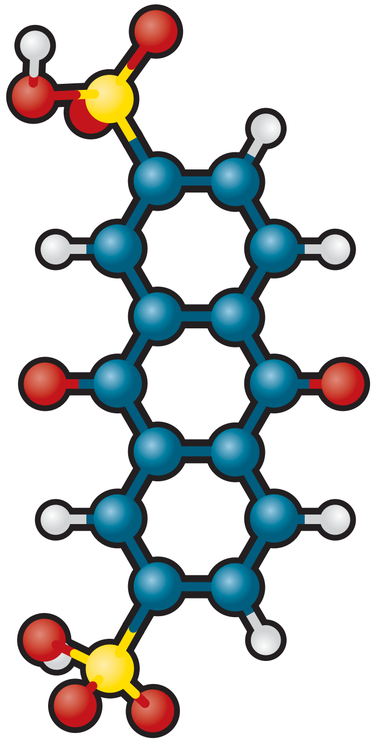
Xie et al JML '22, SciPost Physics '23...

3

Materials design

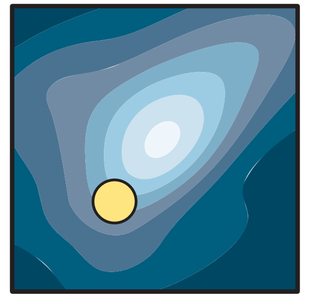
The prediction-generation gap in materials design

Structure
X



Prediction $p(y \mid X)$

Generation $p(X \mid y)$

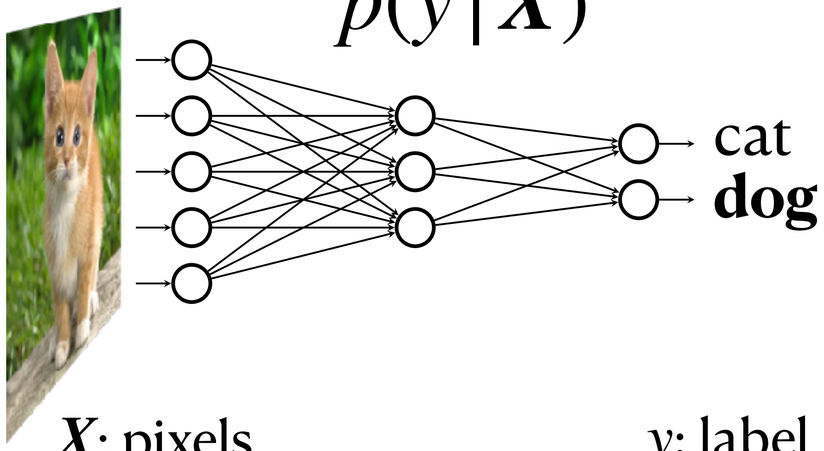


Property
y

Inverse molecular design using machine learning, Sanchez-Lengeling & Aspuru-Guzik, Science '18

Inverse design in search of materials with target functionalities, Zunger, Nature Reviews Chemistry '18

Google's DeepDream

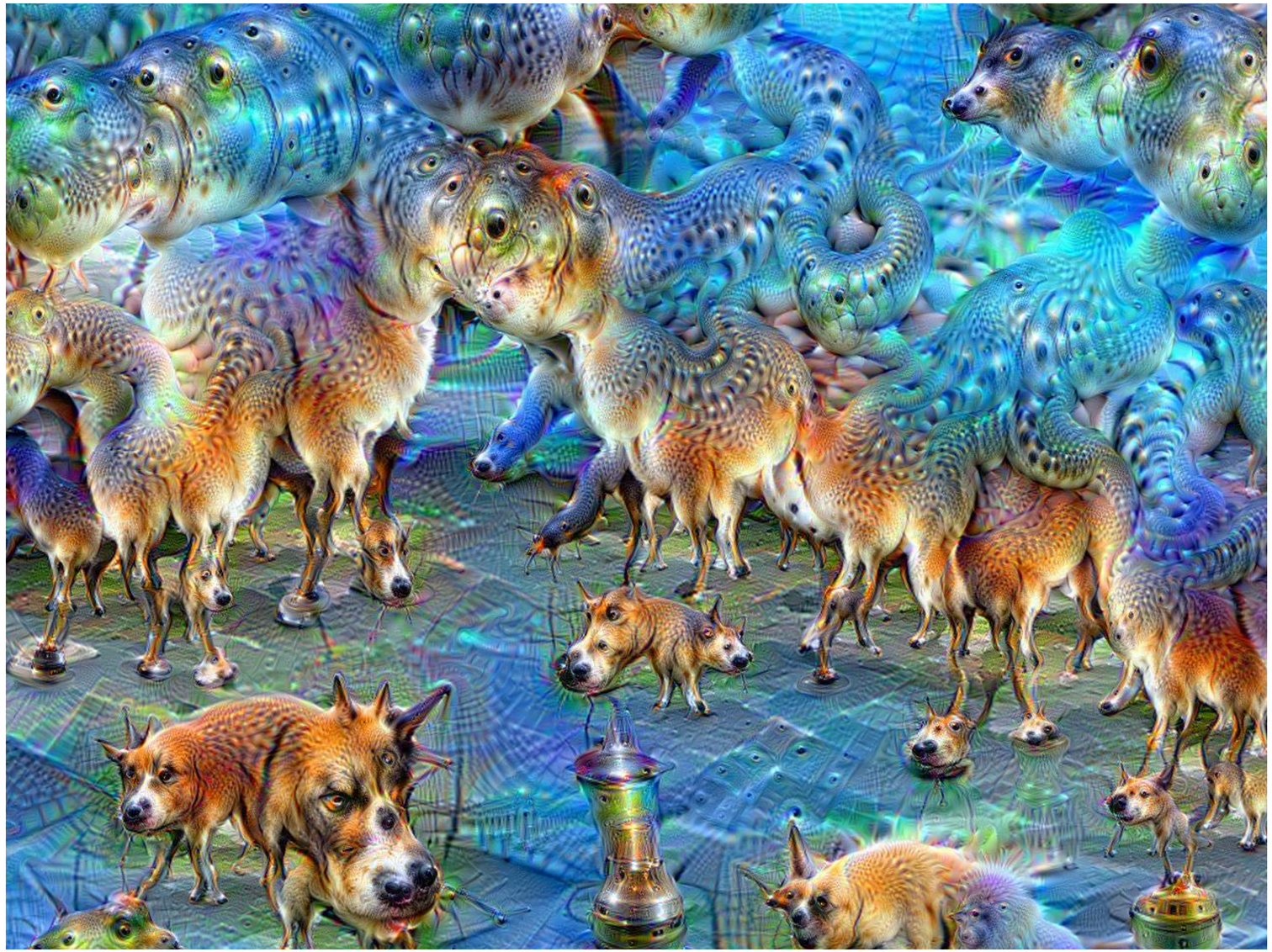


$\boldsymbol{X}$: pixels

y: label

$$\nabla_{\text{pixels}} p(\text{dog} \mid \text{pixels})$$

Backprop a prediction model



http://googleresearch.blogspot.ch/2015/06/inceptionism-going-deeper-into-neural.html

Bayes rule

and the prediction-generation gap

posterior prior likelihood

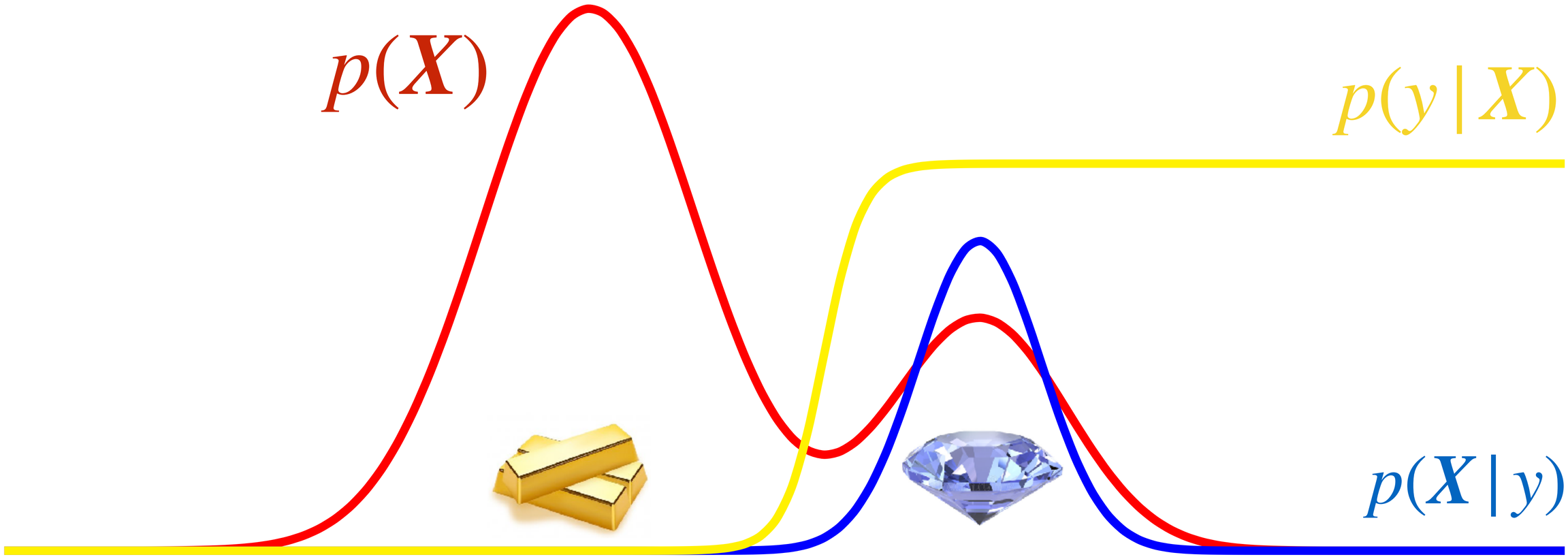
$$p(\boldsymbol{X} \mid y) \propto p(\boldsymbol{X}) p(y \mid \boldsymbol{X})$$

Inverse design Forward prediction

Materials design as posterior sampling

$$\boldsymbol{X} \sim p(\boldsymbol{X} \mid y) \propto p(\boldsymbol{X})\, p(y \mid \boldsymbol{X})$$

Prior Prediction



Variational inference of the posterior

$$\mathbb{KL}\left(q(\boldsymbol{X}) \parallel p(\boldsymbol{X}\,|\,y)\right) = \mathop{\mathbb{E}}_{\boldsymbol{X}\sim q(\boldsymbol{X})}\left[-\ln p(y\,|\,\boldsymbol{X})\right] + \mathbb{KL}\left(q(\boldsymbol{X})\|p(\boldsymbol{X})\right)$$

↑ Variational probability

↑ Likelihood

↑ Relative entropy

Encode prior & regularize the optimization landscape

Ahmed et al, 1811.11214

$q(\boldsymbol{X})$ is easy to sample, e.g. another autoregressive model

Variational inference turns a sampling problem into a stochastic optimization problem



Also known as: reinforcement fine-tuning

$$\mathbb{KL}\left(q(\boldsymbol{X}) \parallel p(\boldsymbol{X}|y)\right) = \mathop{\mathbb{E}}_{\boldsymbol{X}\sim q(\boldsymbol{X})}\left[-r(\boldsymbol{X})\right] + \mathbb{KL}\left(q(\boldsymbol{X})\|p(\boldsymbol{X})\right)$$

↑ Fine-tuned model

↑ Reward function 👍👎

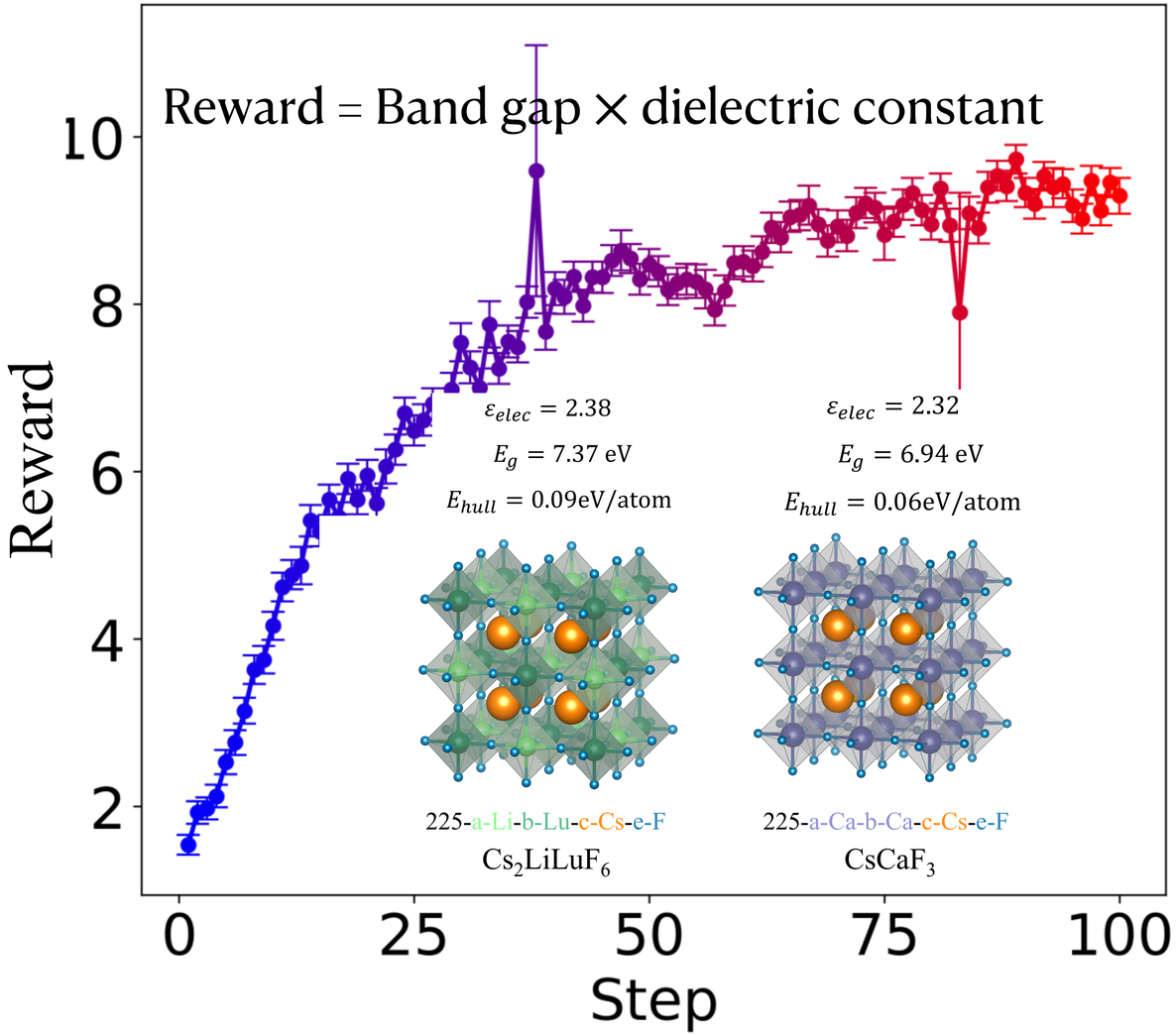
↑ Remain close to the pretrained model

"RL with KL penalties is better viewed as Bayesian inference" Korbak et al, 2205.11275

Reinforcement fine-tuning for materials design

Step 1: pretrain a generative model on known materials

Step 2: fine-tune it on the property reward you care about



$$F = \mathop{\mathbb{E}}_{X\sim q(X)} \left[-r(\boldsymbol{X})\right] + \tau\, \mathbb{KL}\left(q(\boldsymbol{X}) \| p(\boldsymbol{X})\right)$$

Reward

User-specified materials properties

Pretrained prior

Space group informed autoregressive transformer

Cao, Luo, Lv, LW, Sci. Bull. '25

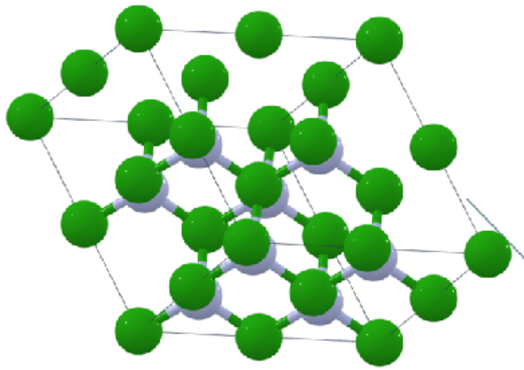
CrystalFormer-RL, Cao, LW, PRB '26

https://github.com/deepmodeling/crystalformer

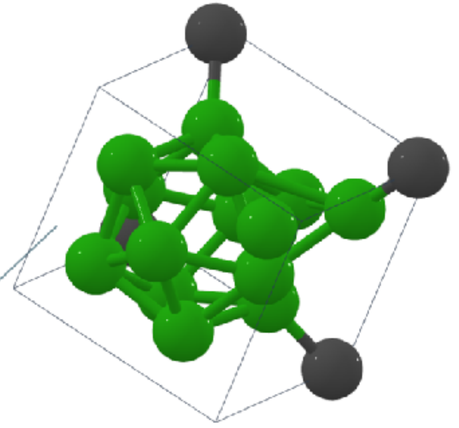
Designing superhard—low mass density materials

ongoing with Kexiang Mao and Xiaohui Yu's group at IOP

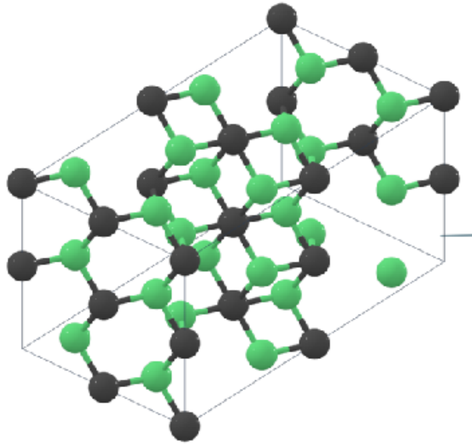
$r(\boldsymbol{X})$ = hardness/density



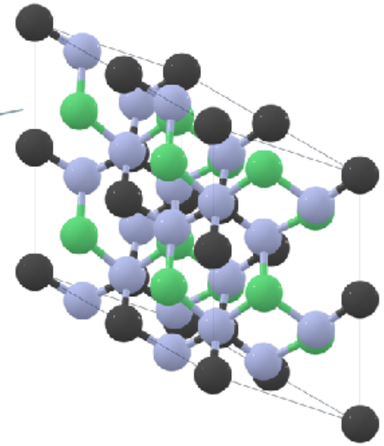
BN



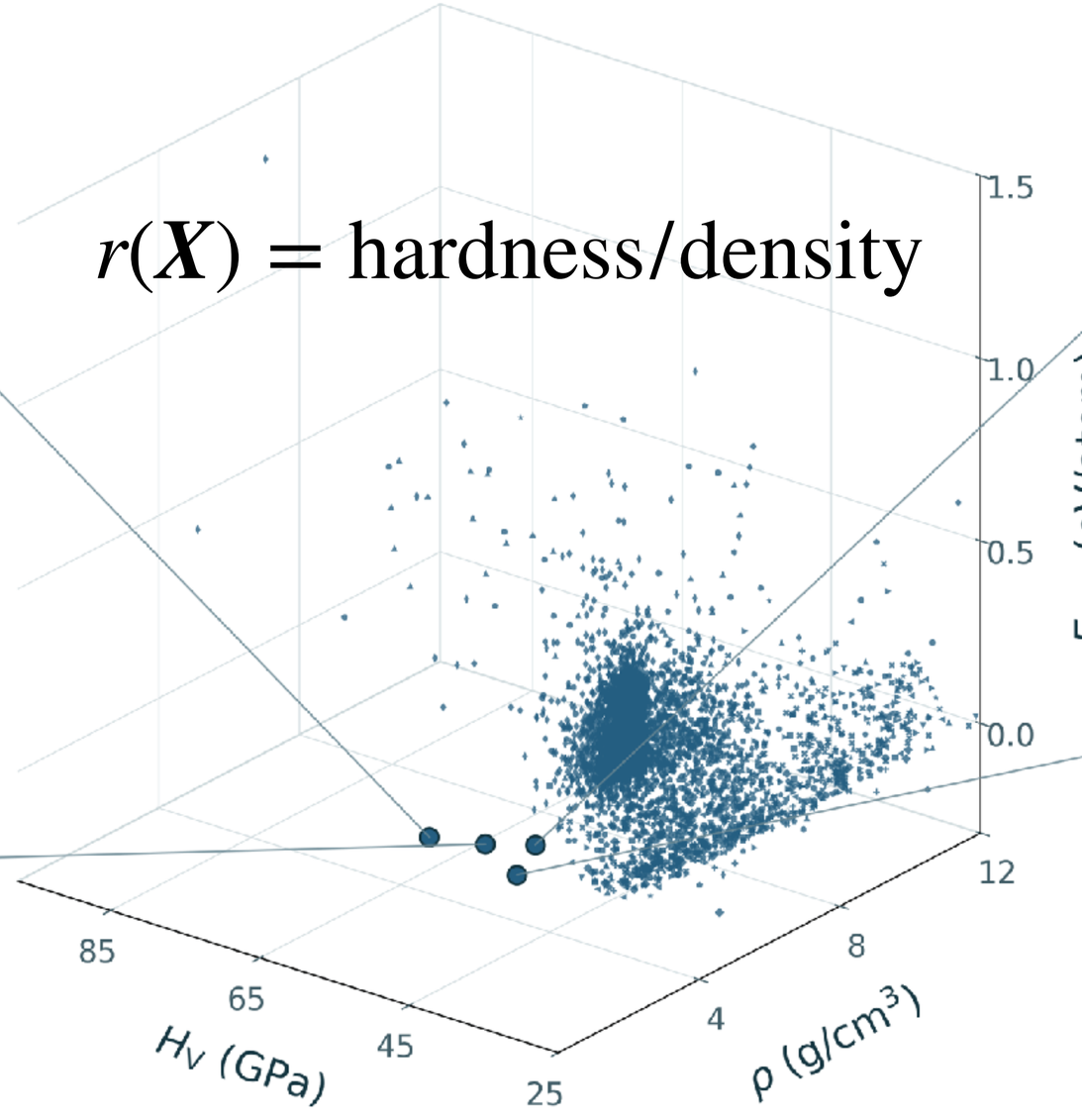
$B_{14}C$



Be_2C



$BeCN_2$



SAME objective, SAME model, SAME algorithm

1

LLM → $X \mapsto r(X)$

LLM post-training

OpenAI, DeepSeek, ...

2

$$F[\rho] = E - TS$$

Nature's cost function

Xie et al JML '22, SciPost Physics '23...

3

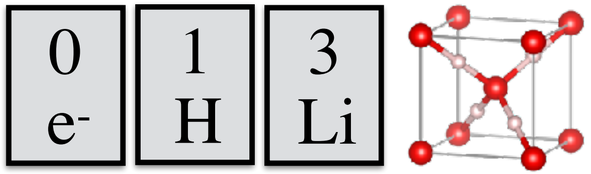
$$p(X \mid y) \propto p(X)p(y \mid X)$$

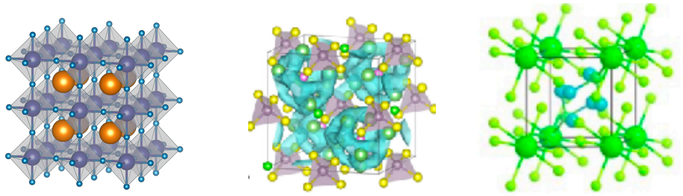
Materials design

Cao et al, 2403.15734 & 2504.02367...



"... the murderer is __"





Generative AI: from Bit to It

"It from Bit"
—Wheeler, 1989